

Given and Taken Away: What a Body Model and a Teacher Leave a Muscle-Driven Learner

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Abstract—Developing learners rarely start from nothing: they have a model of their own body and the guidance of more skilled others, and both kinds of help can be given to an artificial motor learner. Prior work measures such help while it is present. What it leaves behind once withdrawn is unknown, because a withdrawal removes the help’s knowledge and changes the training objective at the same moment. This paper separates the two. One soft actor-critic learner is trained on six muscle-driven bodies, an elbow and a finger from MyoSuite and planar arms of one to four joints, with a frozen body model learned from reward-free babbling and used for imagined rollouts, or with a frozen teacher acting through an imitation term, each against a random counterpart, with twelve or twenty-four matched seeds. A control that replaces the teacher by a frozen copy of the student keeps the objective and removes only the teacher. The body model helps on all six bodies, from a tenth of the error on one joint to nearly half on four, and a random model harms more the more joints there are. A teacher is worth its competence. Withdrawn, neither kind of help leaves the learner detectably worse than one that never had it. On the finger the student keeps 70% of its teacher’s benefit, and the drop at withdrawal is the change of objective, not lost knowledge. The body model’s benefit lasts mainly while its imagination runs.

Index Terms—Developmental robotics, internal models, guidance, musculoskeletal control, reinforcement learning.

I. INTRODUCTION

MUSCLE-DRIVEN bodies are hard to control: each joint is pulled by several muscles, the muscles act with delay, and many activation patterns produce the same movement. Deep reinforcement learning on such bodies needs hundreds of millions of environment steps [1]–[3], while humans and animals acquire new movements in a handful of trials. Computational sensorimotor theory points to two kinds of help a biological learner brings to a task. One is an internal forward model of the body, predicting the consequences of motor commands [4], [5] and learned from self-generated movement. The other is a more skilled other whose guidance steers early attempts. Both are *priors* in the developmental sense, present before task learning begins, and both are easy to give an artificial learner: a model learned from motor babbling supplies imagined experience, and a teacher policy supplies an imitation target.

Help is eventually taken away. A caregiver steps back, a distilled teacher is discarded [6], [7], and a learned model is switched off. Motor-learning research holds that guidance can breed dependence [8], [9], and scaffolding theory judges a tutor by what the learner does after the tutor fades [10]. Existing studies do not settle what help leaves behind. Model-based and teacher–student methods are evaluated while the help is present. When a distillation term is cut, the student

loses the teacher’s knowledge and a term of its objective at the same moment, so a collapse at weaning [7] cannot be attributed to either. Stopping a body model’s imagination likewise changes how much real data each update sees.

The key observation of this paper is that a withdrawal has to be split into what the help knew and how the help entered training. For a teacher, replacing it at withdrawal by a frozen copy of the student keeps the imitation term but carries no further knowledge. The difference between this *self-anchor* and a plain withdrawal prices the term; the difference between the self-anchor and keeping the teacher prices the teacher’s knowledge. For a body model, the learner that never had it must train exactly as the withdrawn learner does after withdrawal. With these controls, and a random counterpart for each kind of help, the two kinds of help are compared on equal terms while they are present and after they are gone.

The study applies these controls to one soft actor-critic learner on six muscle-driven bodies: a one-joint elbow and a four-joint finger from MyoSuite [11], and four planar arms of one to four joints. Fig. 1 shows the central result: a learner whose teacher is withdrawn drops sharply, recovers, and ends no worse than one never guided, and on the finger far ahead of it. The contributions are:

- a controlled comparison of a frozen babble-trained body model and a frozen teacher, each against a random counterpart, showing that the body model helps on all six bodies and helps more the more joints the body has, and that a teacher is worth its competence (Sec. IV-A–IV-B);
- evidence that withdrawing a teacher leaves no detectable dependence: the withdrawn learner keeps 70% of the teacher’s benefit on the finger, and on the elbow any deficit stays within a quarter of the teacher’s worth after 200k steps (Sec. IV-C);
- a decomposition showing that the drop at withdrawal is the change of objective rather than lost knowledge, and that a body model’s benefit lasts mainly while its imagination runs (Sec. IV-C–IV-D).

Code and data will be released on acceptance.

II. RELATED WORK

Guidance and dependence. The guidance hypothesis [8], [9], [12] holds that frequent augmented feedback guides performance during acquisition but degrades learning. High-frequency physical guidance gives the poorest retention in one test [13], and reduced knowledge of results improves delayed

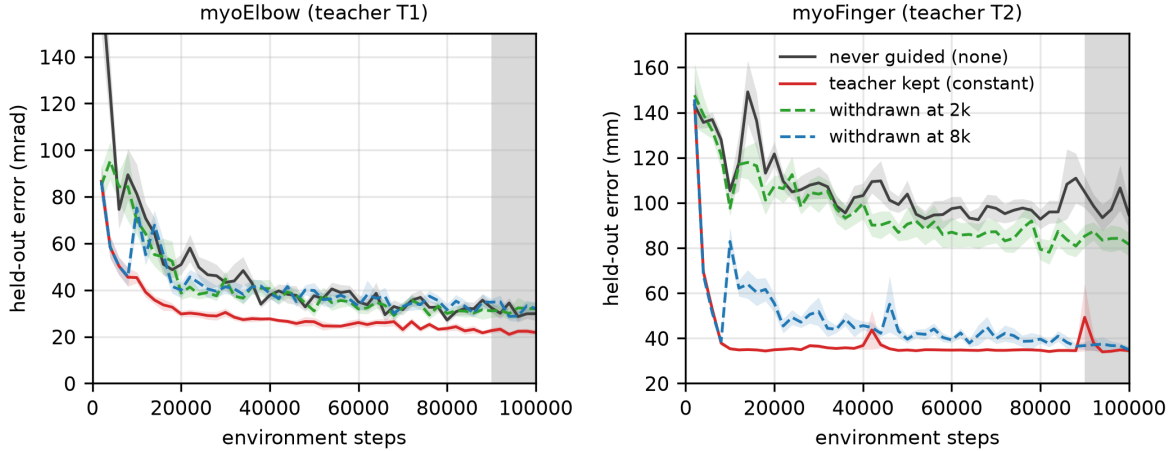


Fig. 1. Taking the teacher away. Held-out error of a learner never guided (grey), a learner that keeps its teacher (red), and learners whose teacher is withdrawn at 2k or 8k steps (dashed). Withdrawal is followed by a sharp drop; the withdrawn learner then recovers. On the finger (right) it ends far ahead of the never-guided learner, more so the longer it was guided; on the elbow (left) it ends level with it. Mean across 12 seeds, band ± 1 SE; the grey band is the window scored as the endpoint.

retention in another [14]. The evidence is mixed: a meta-analysis questions whether frequent feedback degrades learning at all [15], a later one finds no reliable benefit of reduced feedback frequency [16], and children given reduced feedback learn worse, not better [17]. These studies vary how *often* feedback is given and test retention without further practice. The present study varies how *long* a teacher is attached and measures the learner after continued practice; it borrows the question, whether guidance leaves a learner worse off without it, and does not test the hypothesis itself. Human studies corrupt or remove the information in a feedback channel while keeping the channel [18]–[20], and models of dependence on robotic assistance place it in an effort or slacking term of the learner’s own cost [21], [22]. None separates the information from the channel at the moment guidance is withdrawn.

Teacher–student learning. Privileged learning distills a teacher that sees state the deployed policy will not into a student trained on its own rollouts [23]–[25]; RMA [26] regresses the privileged latent instead. Distillation is either the student’s entire training or an auxiliary term annealed or cut to make the student independent [6], [7]. Weaning can collapse the student, which Agarwal et al. [7] attribute, without a test, to the change of objective. A term toward the learner’s own earlier policy stabilises fine-tuning [27], [28]. Supervised distillation uses the student itself or a random network as the teacher to separate what the term does from what the teacher knows [29], [30], and compares stopping distillation mid-way only with continuing it [31]; in language models a teacher-free signal matches a noisy teacher [32]. A cloned policy fine-tuned by off-policy reinforcement often degrades at the transition, because a fresh critic gives unreliable values outside the data [33]–[35]; a retained cloning term prevents this [36]. What is missing is a comparison of the withdrawn student with one never taught, with the change of objective held apart.

Body models. Developmental robotics [37]–[39] treats motor babbling as a way to learn forward models of the self [40], intrinsic motivation as what drives it [41], and infant reaching

as the paradigm case [42]. In model-based reinforcement learning, World Models [43] trains a model once from random rollouts; the Dreamer series [44]–[46] and TD-MPC2 [47] keep the model learning online; MBPO [48] branches short rollouts from real states. Self-modelling [49] and goal babbling [50] learn body models by directed exploration; Plan2Explore [51] and APV [52] pretrain a model without task reward; offline methods [53]–[55] fit a model to logged, rewarded data. Most muscle reinforcement learning is model-free [1]–[3], [56]; muscle world models are co-learned with the skill [57], [58], and frozen models learned from reward-free actuation drive soft and muscle-driven bodies by planning or model-only learning [59]–[61]. An inaccurate model makes model-based learning worse than model-free, more so in higher dimension [48], [62]. These studies price a model while it is in use; how its worth scales with the body, and what remains when it is removed, are open.

III. METHOD

A. Bodies, task and backbone

`myoElbowPose1D6M` is a one-degree-of-freedom elbow driven by six muscles; error is a joint-angle error in milliradians (mrad). `myoFingerReachRandom-v0` is a four-joint index finger driven by five muscles reaching a three-dimensional fingertip target; error is a Euclidean fingertip error in millimetres (mm). Four planar arms of one, two, three and four revolute joints, driven by two, four, eight and ten MuJoCo muscles (antagonist pairs, biarticular from three joints on), reach a planar fingertip target; error is in mm. Errors are never compared across bodies. Episodes are 100 steps from a fixed start pose; training draws from eight fixed targets; evaluation is greedy on twenty disjoint held-out targets, so every number is a generalisation measurement (on the one-joint bodies an interpolation test). One soft actor-critic (SAC) [63], [64] backbone is used in every condition; conditions differ only in what is added.

B. The two kinds of help

The *body model* is a forward model $f_\theta(s, a) \mapsto s'$ trained once on task-agnostic, reward-free Ornstein–Uhlenbeck muscle activations (10^5 transitions on the elbow, 2×10^5 on the finger and on each planar arm), then frozen, and used for short-horizon Dyna [65] rollouts branched from sampled real states. The reward at an imagined state is the environment’s analytic reach reward, computed with the simulator’s forward kinematics, so this route is scoped to a forward model *plus* a known reward and fingertip map. Each update draws 64 real and 64 imagined transitions where model-free conditions draw 128 real.

The *teacher* is a frozen actor. The student’s actor loss is

$$\mathcal{L}_\pi = q \mathcal{L}_{\text{SAC}} + \frac{c}{|\mathcal{A}|} \left\| \tanh \mu_{\text{student}}(o) - \tanh \mu_{\text{teacher}}(o) \right\|^2, \quad (1)$$

on replayed observations o , where $|\mathcal{A}|$ is the number of muscles and $c = 1$ while the teacher is attached; the critic trains on reward in every condition. In the *additive* regime ($q = 1$ always) the imitation term supplements the student’s own policy gradient; this is the teacher used everywhere unless stated. In the *replacing* regime ($q = 0$ while attached, $q = 1$ from withdrawal) only the teacher moves the actor until withdrawal, as in reward-free distillation [24], [25].

Two teachers are used. T1 on each MyoSuite body is the terminal actor of a model-free run of 30k rewarded steps on a seed outside the study’s twelve; it scores 27.2 mrad on the elbow’s held-out targets, and 169.1 mm on the finger’s, where a zero-activation policy scores 172.3, so the finger T1 barely moves. T2 on the finger is chosen to be competent: it is the terminal actor of the best of the twelve body-model runs of 100k steps (35.6 mm held-out; the twelve range from 35.6 to 85.3, mean 54.4), and it is the teacher of every finger withdrawal experiment. One of the twelve students shares its initialisation seed with the run that produced T2; excluding it changes no finger-T2 conclusion.

C. Conditions

Worth of the help. `blank` is the backbone alone; `prior` adds the body model; `coach` adds the teacher. `randprior` and `randcoach` replace the model or the teacher by a randomly initialised one in identical machinery (on the one- and four-joint arms the random model carries the input normalisers of an earlier body model, which differ from the final one’s by under 1.5%). `blank64` builds every update from 64 distinct real transitions, each repeated once to fill the batch: the real-data content of a Dyna update and no model (SAC at batch 64, up to the independent action samples drawn for the repeated rows). `priorcoach` carries both kinds of help.

Taking the teacher away at step t_w : `none` (never guided, identical to `blank`); `constant` (teacher kept, identical to `coach`); `abrupt` ($c \rightarrow 0$ at t_w); `selfanchor` (the teacher is replaced by a frozen copy of the student’s own actor, c unchanged); `randanchor` (replaced by a freshly initialised actor); and, in the replacing regime only, `gateon` ($q \rightarrow 1$ at t_w with the teacher kept). The anchor swaps fire two environment steps before t_w , at the step the teacher’s weight

is last written. The withdrawn learner is compared with two references: with `none`, to ask whether it is worse off than a learner never guided, and with `constant`, to ask what it gives up against keeping the teacher. The self-anchor splits the second: `selfanchor–constant` is losing the teacher with the imitation term kept (the *teacher* component), `abrupt–selfanchor` is losing the term (the *loss-term* component). The snapshot holds whatever of the teacher the student has absorbed by t_w , so the teacher component measures only what the teacher still has to add. Since the snapshot’s gradient is zero at the swap, the loss-term component is the value of keeping a stabilising term after withdrawal, not an instantaneous shock.

Taking the body model away at step 8000: `keep` (identical to `prior`); `purge` stops imagination and empties the imagined buffer, after which every batch is 128 real, as in `blank`; `matched withdraws` as `purge` does but draws 64 real transitions and repeats them, as in `blank64`. Each withdrawn condition is compared with the never-had condition that trains as it does after withdrawal: `purge` with `blank`, `matched` with `blank64`.

D. Metrics and statistics

Endpoint: the mean of a run’s six held-out evaluations at 90k, 92k, ..., 100k (the *late mean*); single evaluations of one run move by several mrad or mm between neighbouring checkpoints. The late mean replaced a single 100k evaluation after an internal review. Under the single evaluation, 21 of the endpoint contrasts reported here change detection, among them the elbow body model’s benefit over `blank` and the elbow shortfall after the model is withdrawn (neither detected at 100k alone). The pooled withdrawn-versus-never-guided contrasts keep their detection, but the elbow’s one-sided bound on a deficit would be 10.76 mrad, above the teacher’s worth, so the non-inferiority statement on that body rests on the late mean and on the longer runs of Sec. IV-C. **Dip:** error at $t_w + 1000$ minus error at t_w . **Attachment trend:** per-seed ordinary-least-squares slope of a contrast against t_w , per 1,000 attached steps.

The seed is the unit of analysis; every condition has $n = 12$ matched seeds except the three- and four-joint arms, which have $n = 24$; every comparison is a paired contrast on the seed; intervals are paired- t 95% intervals, and an interval excluding zero is marked * or set in bold. A non-detected contrast is not evidence of no difference. Where a withdrawn learner is said not to be worse off, the one-sided 95% upper bound of its deficit is given and compared with what the help was worth. Significant cells of Tables I and IV were checked with an exact Wilcoxon signed-rank test; the one it does not confirm at $p < 0.05$ is flagged ‡. Convergence is checked per condition as the paired change from the 70–80k window to the 90–100k window: 28 of the 97 conditions behind the results are still improving or worsening (4.9 expected by chance), while none of the four pooled teacher-withdrawal contrasts of Table III is (elbow, additive: $+1.52 [-1.96, +5.00]$ mrad).

E. Implementation details

Actor and critics are two-layer 128-unit SiLU networks, the twin critics with layer normalisation; $\gamma = 0.99$, Polyak

$\tau = 0.005$, Adam at 3×10^{-4} , target entropy $-0.5 |A|$, gradient norm clipped at 10, real replay 200,000 transitions, imagined replay 400,000, 1,000 warm-up steps, update-to-data ratio 2, batch 128. Dyna: every 100 environment steps, 200 start states are drawn from real replay and rolled $k = 3$ steps under the current actor through the frozen model. The forward model is a two-layer 256-unit SiLU network predicting the next state, trained with Adam at 10^{-3} for 40 epochs on babble data ($\theta = 0.15$, $\sigma = 0.4$, clipped to $[-1, 1]$, pose re-randomised every 100 steps, no reward). Teachers have the actor's architecture. Evaluation is one greedy 100-step episode on each of the twenty held-out targets, every 2,000 steps, and every 1,000 in the withdrawal conditions. A matched seed sets the network initialisation, replay sampling, the training environment and the order of target draws; the targets are fixed across seeds, and runs are deterministic on CPU. The body model sees more transitions than T1 was trained on; rebuilding the model from exactly 30k babble transitions changes its student's error at 12k by $+2.75 [-12.77, +18.27]$ mm on the finger and $+1.55 [-0.87, +3.97]$ mrad on the elbow, neither detected. The 1,608 runs behind the results take about 1,700 single-thread CPU-hours.

IV. RESULTS

A. The body model helps, more the more joints there are

The body model beats model-free learning on both MyoSuite bodies (Table I, Fig. 2): by $-3.90 [-7.42, -0.38]$ mrad on the elbow and $-46.19 [-58.70, -33.69]$ mm on the finger. Part of a Dyna update's effect could come from its smaller real batch alone, since smaller batches can help off-policy learners by reducing critic overfitting [66]–[68]; on the finger halving the real transitions per update does help ($-20.29 [-32.53, -8.05]$). Against this `blank64` control the model still wins on both bodies, by $-25.90 [-33.77, -18.04]$ mm on the finger. The model's worth is its content: a random model in the same machinery is worse than the trained one on both bodies, and worse than no model on the elbow ($+21.88 [+14.60, +29.15]$).

The planar arms show how this worth scales with the body (Table II, Fig. 3). The model beats model-free learning on every arm, by 10%, 18%, 20% and 47% of the model-free error on one to four joints; the per-seed slope of this relative benefit is $-0.10 [-0.19, -0.01]$ per joint (exact Wilcoxon $p = 0.034$). The more joints there are to coordinate, the more the imagined experience is worth. A random model runs the other way: harmless or slightly helpful on one joint ($-4.24 [-6.93, -1.54]$ mm, the one body where it is not detectably worse than the trained model) and increasingly harmful after, $+262.20 [+198.84, +325.57]$ mm on four joints; its relative harm grows by $+0.44 [+0.30, +0.58]$ of the model-free error per joint ($p < 0.001$). Against `blank64` the trained model is ahead on the one-, two- and four-joint arms and not detectably on the three-joint arm. Over the six bodies the babble model beats model-free learning on six, `blank64` on five and the random model on five.

B. A teacher is worth its competence

On the elbow the teacher T1 leaves its student $-8.65 [-12.25, -5.05]$ mrad ahead of model-free learning and $-4.75 [-6.28, -3.22]$ ahead of the body model. On the finger the teacher decides the outcome. T1, which barely moves, holds its student $+21.36 [+9.83, +32.89]$ mm above model-free learning, not detectably different from a student held near a random teacher. T2 on the same body and seeds gives a student at 37.5 mm, near its teacher's 35.6: $-61.71 [-74.98, -48.44]$ below model-free learning and $-15.52 [-24.43, -6.60]$ below the body model. Adding the body model to a teacher changes little (Table I, last rows). Which kind of help is worth more is therefore a property of the teacher, not of the route.

C. Taking the teacher away leaves no dependence

The teacher is withdrawn at every 1,000 steps from 2,000 to 8,000, and each withdrawn learner trains on to 100k (Table III, Fig. 1). On the finger the withdrawn learner ends $-43.25 [-54.72, -31.79]$ mm ahead of one never guided, pooled over the seven withdrawal times: it keeps 70% of the teacher's worth. It keeps more the longer the teacher was attached, from $-14.82 [-27.27, -2.36]$ at $t_w = 2000$ to -62.60 at $t_w = 8000$ (slope $-8.70 [-10.07, -7.33]$ per 1,000 attached steps), and from $t_w = 7000$ on it is not detectably behind a learner that kept its teacher (Table V, total). On the elbow the withdrawn learner is not detectably different from one never guided at any t_w ($+1.76 [-1.36, +4.89]$ pooled, no trend); any deficit is at most 4.31 mrad at 95% confidence, under half of the 8.65 mrad the teacher was worth. On this body the teacher's advantage does not outlast it: the withdrawn learner is 9 to 12 mrad behind one that kept the teacher at every t_w .

Training twice as long tightens the elbow result. The conditions none, constant and abrupt at all seven withdrawal times run to 200k on the same seeds; their first 100k reproduce the runs above at every checkpoint. Scored over 190–200k, the withdrawn learner is $+0.04 [-2.11, +2.19]$ mrad from one never guided, with a one-sided bound of 1.79 mrad against a teacher worth $-7.50 [-10.18, -4.82]$; no single withdrawal time is detected, and the gap to a learner that kept its teacher persists (5.7 to 9.9 mrad, detected at every t_w). The never-guided learner is still improving ($-2.79 [-4.99, -0.60]$ from the 170–180k window), and the pooled contrast moves with it, from $-2.32 [-5.25, +0.61]$ at 170–180k to level.

When the teacher replaces the student's own policy gradient until withdrawal, the outcome is the same. Withdrawn at 2k, 4k, 6k or 8k, the finger learner ends $-55.87 [-66.36, -45.39]$ mm ahead of one never guided (89% of the teacher's worth) and the elbow learner is not detectably different from it ($+1.09 [-2.64, +4.82]$). On the elbow the pure-imitation student is only $-4.01 [-6.57, -1.44]$ ahead of model-free learning even with its teacher kept, so this cell cannot bound a deficit below the teacher's worth. The transition is harder in this regime: four thousand steps after withdrawal the elbow learner is 80 to 202 mrad behind one that kept its teacher. A learner that keeps its teacher but

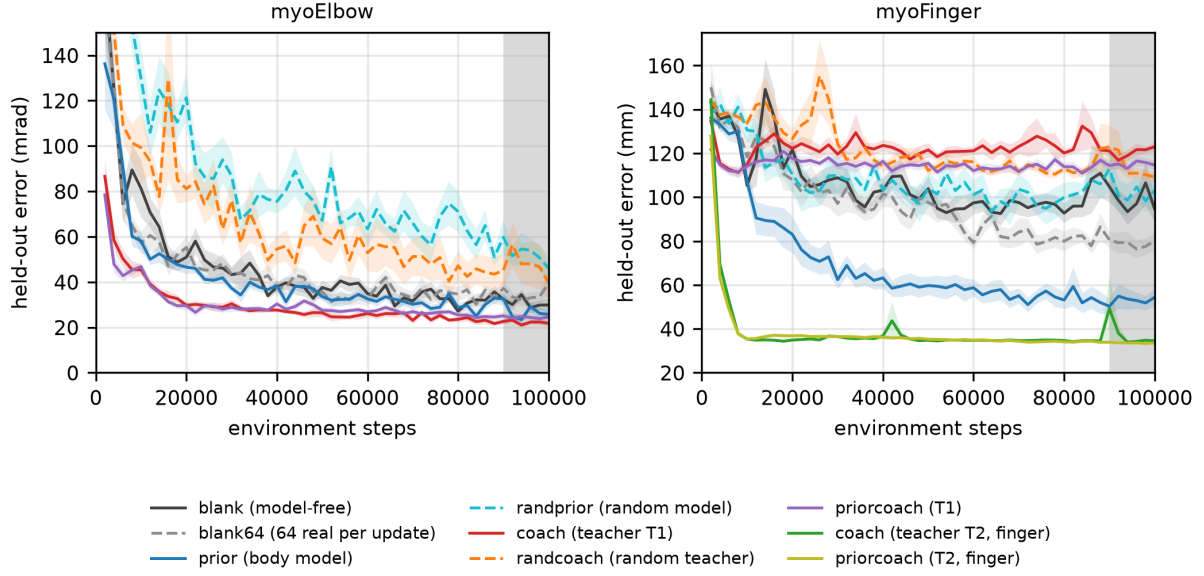


Fig. 2. What each kind of help is worth on the two MyoSuite bodies. The body model (blue) ends below model-free learning (black) on both; a random model (cyan, dashed) is worse than the trained one. A teacher is worth its competence: on the finger the near-motionless teacher T1 (red) holds its student above model-free learning, and the competent teacher T2 (green) gives the best condition. Held-out error, mean across 12 seeds, band ± 1 SE; elbow axis clipped at 150 mrad; grey band: the scored window.

TABLE I

WHAT THE HELP IS WORTH ON THE TWO MYOSUITE BODIES: LATE-MEAN ERROR (90–100K), $n = 12$. MEANS, THEN PAIRED CONTRASTS WITH 95% INTERVALS; NEGATIVE FAVOURS THE FIRST-NAMED CONDITION; BOLD MARKS AN INTERVAL EXCLUDING ZERO. FINGER TEACHER CONDITIONS ARE GIVEN FOR T1 (BARELY MOVES) AND T2 (COMPETENT). ‡: EXACT WILCOXON $p \geq 0.05$.

	myoElbow (mrad)	myoFinger, T1 (mm)	myoFinger, T2 (mm)
blank / blank64	30.88 / 34.80	99.17 / 78.88	
prior / randprior	26.98 / 52.75	52.98 / 103.09	
coach / randcoach	22.23 / 47.29	120.53 / 114.25	37.46 / 114.25
priorcoach	24.46	114.94	33.59
prior-blank	-3.90 [-7.42, -0.38]	-46.19 [-58.70, -33.69]	
prior-blank64	-7.82 [-11.02, -4.62]	-25.90 [-33.77, -18.04]	
blank64-blank	+3.92 [+1.26, +6.58]	-20.29 [-32.53, -8.05]	
prior-randprior	-25.78 [-34.71, -16.84]	-50.11 [-64.54, -35.68]	
randprior-blank	+21.88 [+14.60, +29.15]	+3.92 [-14.49, +22.34]	
coach-blank	-8.65 [-12.25, -5.05]	+21.36 [+9.83, +32.89]	-61.71 [-74.98, -48.44]
coach-prior	-4.75 [-6.28, -3.22]	+67.55 [+58.80, +76.29]	-15.52 [-24.43, -6.60]
coach-randcoach	-25.06 [-40.64, -9.48]	+6.28 [-2.71, +15.27]	-76.79 [-87.26, -66.31]
priorcoach-coach	+2.23 [-1.04, +5.50]	-5.58 ‡ [-10.95, -0.21]	-3.87 [-11.06, +3.32]
coachT2-coachT1		-83.07 [-89.88, -76.25]	

TABLE II

JOINT-COUNT SERIES: FOUR PLANAR MUSCLE ARMS, LATE-MEAN ERROR IN MM. MEANS, THEN PAIRED CONTRASTS WITH 95% INTERVALS ($df = n - 1$); BOLD MARKS AN INTERVAL EXCLUDING ZERO. THE MODEL'S BENEFIT GROWS FROM 10% TO 47% OF THE MODEL-FREE ERROR WITH JOINT COUNT; A RANDOM MODEL'S HARM GROWS FASTER.

	1 joint ($n=12$)	2 joints ($n=12$)	3 joints ($n=24$)	4 joints ($n=24$)
blank	93.4	34.9	176.7	200.6
blank64	96.0	38.3	149.0	140.9
prior	84.3	28.5	141.2	105.6
randprior	89.2	56.9	274.1	462.8
prior-blank	-9.16 [-14.98, -3.34]	-6.40 [-11.09, -1.71]	-35.52 [-62.29, -8.75]	-94.99 [-134.00, -55.98]
as % of blank	-10%	-18%	-20%	-47%
prior-blank64	-11.74 [-21.81, -1.66]	-9.77 [-16.65, -2.89]	-7.84 [-30.11, +14.44]	-35.28 [-59.50, -11.06]
blank64-blank	+2.58 [-4.82, +9.98]	+3.37 [-3.61, +10.35]	-27.68 [-56.25, +0.88]	-59.72 [-102.93, -16.51]
randprior-blank	-4.24 [-6.93, -1.54]	+22.02 [+9.81, +34.23]	+97.37 [+41.17, +153.56]	+262.20 [+198.84, +325.57]
prior-randprior	-4.92 [-10.28, +0.43]	-28.42 [-41.59, -15.25]	-132.89 [-183.14, -82.64]	-357.20 [-406.71, -307.69]

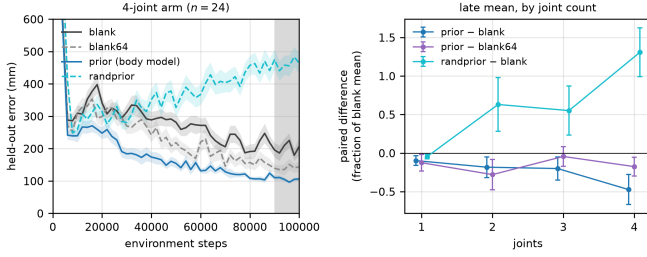


Fig. 3. The body model is worth more on harder bodies. Left: the four-joint arm over training (mean, ± 1 SE, $n = 24$; grey band: the scored window); the body model (blue) ends at half the model-free error (black), the random model (cyan) far above it. Right: late-mean paired contrasts as a fraction of each arm’s model-free error, with 95% intervals.

TABLE III

TAKING THE TEACHER AWAY: THE WITHDRAWN LEARNER AGAINST ONE NEVER GUIDED (NONE), POOLED OVER THE n_{t_w} WITHDRAWAL TIMES TESTED, LATE MEAN, $n = 12$. UPPER: ONE-SIDED 95% BOUND OF THE MEAN. WORTH: WHAT KEEPING THE TEACHER GIVES OVER NO TEACHER (CONSTANT-NONE). EVERY DEFICIT BOUND LIES BELOW THE TEACHER’S WORTH, EXCEPT ON THE ELBOW IN THE REPLACING REGIME, WHERE THE WORTH ITSELF IS SMALL. ELBOW IN MRAD (T1), FINGER IN MM (T2).

Body, regime	n_{t_w}	withdrawn – none	upper	worth
Elbow, additive	7	+1.76 [–1.36, +4.89]	+4.31	–8.65
Finger, additive	7	–43.25 [–54.72, –31.79]	–33.90	–61.71
Elbow, replacing	4	+1.09 [–2.64, +4.82]	+4.14	–4.01
Finger, replacing	4	–55.87 [–66.36, –45.39]	–47.32	–62.91

TABLE IV

THE DROP AT WITHDRAWAL IS THE LOSS TERM. COST OF TAKING THE TEACHER AWAY AT $t_w = 8000$, DECOMPOSED, $n = 12$: IMMEDIATE DIP, THEN THE SAME CONTRASTS AT 12K AND AT THE LATE MEAN. ROWS 1–3 ARE THE TOTAL, THE TEACHER COMPONENT AND THE LOSS-TERM COMPONENT. * MARKS AN INTERVAL EXCLUDING ZERO. ELBOW IN MRAD (TEACHER T1), FINGER IN MM (TEACHER T2).

Contrast	myoElbow			myoFinger, T2		
	dip	12k	end	dip	12k	end
abrupt – constant	+45.17*	+16.00*	+9.90*	+49.20*	+27.34*	–0.89
selfanchor – constant	+4.95	+9.42*	+8.61*	+2.97*	+3.61*	–2.44
abrupt – selfanchor	+40.22*	+6.59	+1.29	+46.22*	+23.72*	+1.55
randanchor – selfanchor	+47.44*	+20.66*	+2.55	+70.47*	+85.93*	+73.79*
abrupt – none	+43.64*	–15.60	+1.25	+55.03*	–55.43*	–62.60*
constant – none	–	–31.61*	–8.65*	–	–82.77*	–61.71*

switches on its own policy gradient (gateon) also falls, by 25 to 85 mrad, on average as far as one whose teacher is replaced by its own snapshot. Much of the cost is therefore the first use of the student’s own gradient after an imitation-only phase, the known collapse when a cloned policy is fine-tuned by off-policy reinforcement [33]–[35]. By the end of training the withdrawn learner is not detectably different from a never-guided one.

The drop at withdrawal is the change of objective. Withdrawal is followed by a sharp drop (Fig. 1): at $t_w = 8000$ the error rises by +45.17 [+24.58, +65.76] mrad on the elbow and +49.20 [+31.79, +66.60] mm on the finger over the next 1,000 steps, relative to keeping the teacher (Table IV). The self-anchor, which keeps the imitation term and removes only the teacher, removes most of

the drop: the teacher component is +4.95 [–1.24, +11.15] on the elbow and +2.97 [+0.24, +5.70] on the finger, and the loss-term component is +40.22 [+19.96, +60.48] and +46.22 [+30.30, +62.15]. A random anchor in the same place is far worse than the self-anchor on both bodies, so the control works because its target is where the student already points, not because any term will do. Across the attachment sweep (Table V, Fig. 4) the teacher component of the drop falls the longer the teacher was attached (–6.23 [–7.79, –4.68] mrad and –4.65 [–7.05, –2.25] mm per 1,000 steps): the longer the attachment, the more of the teacher the student has absorbed, and the less the teacher still has to add. The loss-term component does not fall; on the finger it grows (+7.07 [+2.95, +11.19]). The finger’s drop against the never-guided learner also grows with attachment (+9.34 [+2.07, +16.61]), but most of that growth is the never-guided learner’s own progress over the same window, which changes with t_w (+6.92 [+2.34, +11.50]). By the end of training the loss-term component is not detected at $t_w = 8000$ on either body; on the elbow what the withdrawn learner gives up against a kept teacher is the teacher component, +8.61 [+4.18, +13.04], the size of the teacher’s worth.

D. A body model’s benefit lasts mainly while it runs

A body model helps through imagination, so the question is how much of its benefit survives once imagination stops (Table VI). On the finger, about half: withdrawn at 8k, the learner ends –26.05 [–38.06, –14.04] mm ahead of a never-had learner that trains as it does (purge against blank, 56% of the model’s worth), and –11.44 [–22.33, –0.56] ahead with real data per update held at 64 (matched against blank64, 44%). On the elbow, where the model is worth 3.90 mrad, neither withdrawn condition is distinguishable from a never-had learner. On the four-joint arm, where it is worth 95 mm, none of the benefit is detected after withdrawal (+5.84 [–51.63, +63.31] and +10.71 [–18.86, +40.28]), though the purge learner is still improving at 100k (–48.57 [–72.23, –24.92] over the last 20k steps) while blank is not. Against blank instead of blank64, the matched learner on this arm would appear 49 mm ahead; that difference is the smaller real batch, which on this body helps by itself. On every body the withdrawn learner falls short of one that kept its model, by +7.13 [+3.18, +11.07] mrad, +14.46 [+6.63, +22.29] mm and +45.99 [+25.51, +66.47] mm.

V. DISCUSSION

A lesson and a tool. The two kinds of help leave different things behind. A teacher’s help is written into the policy: on the finger the student that loses its teacher keeps most of what the teacher gave, more the longer it was taught, and the self-anchor shows that what the teacher still has to add at withdrawal shrinks as the student absorbs it. A body model’s help is a way of learning: it supplies imagined experience while it runs, is worth more the more joints there are to coordinate, and is worth nothing, or harmful, when its content is random. When imagination stops, part of what it bought

TABLE V

TEACHER-WITHDRAWAL COMPONENTS AGAINST ATTACHMENT DURATION t_w : THE IMMEDIATE DIP, AND THE LATE MEAN. CELLS ARE PAIRED MEANS, $n = 12$, * AN INTERVAL EXCLUDING ZERO; SLOPE: PER-SEED OLS OVER t_w , PAIRED- t INTERVAL, PER 1,000 ATTACHED STEPS. THE DIP OF ABRUPT — NONE IS THE WITHDRAWN LEARNER'S DIP MINUS THE NEVER-GUIDED LEARNER'S CHANGE OVER THE SAME 1,000 STEPS.

		2k	3k	4k	5k	6k	7k	8k	slope per 1k
<i>myoElbow (mrad), teacher T1</i>									
dip	teacher	+45.97*	+22.02*	+25.64*	+1.52	+2.68	+7.76	+4.95	-6.23 [-7.79, -4.68]*
	loss term	+13.55	+17.65*	+37.91*	+42.57	+77.88*	+24.29*	+40.22*	+4.76 [-0.13, +9.65]
	total	+59.52*	+39.67*	+63.55*	+44.09*	+80.56*	+32.05*	+45.17*	-1.47 [-5.79, +2.84]
	abrupt — none	+55.44	+63.67*	+74.68*	+75.04*	+79.60*	+13.43	+43.64*	-4.68 [-14.80, +5.44]
end	teacher	+3.51	+7.48*	+2.84	+3.69	+7.54*	+8.27*	+8.61*	+0.77 [-0.55, +2.09]
	loss term	+5.73	+2.97	+7.26*	+5.92*	+3.90	+3.89	+1.29	-0.53 [-2.05, +0.99]
	total	+9.24*	+10.45*	+10.10*	+9.61*	+11.44*	+12.16*	+9.90*	+0.24 [-0.36, +0.84]
	abrupt — none	+0.59	+1.80	+1.45	+0.96	+2.79	+3.51	+1.25	+0.24 [-0.36, +0.84]
<i>myoFinger (mm), teacher T2</i>									
dip	teacher	+35.69*	+23.60*	+17.21*	+21.44*	+16.15*	+8.13*	+2.97*	-4.65 [-7.05, -2.25]*
	loss term	+7.29	+22.05	+40.25*	+39.31*	+56.39*	+54.61*	+46.22*	+7.07 [+2.95, +11.19]*
	total	+42.98*	+45.65*	+57.46*	+60.75*	+72.54*	+62.74*	+49.20*	+2.42 [-2.41, +7.26]
	abrupt — none	-7.34	+28.62*	+53.89*	+46.47*	+72.82*	+56.39*	+55.03*	+9.34 [+2.07, +16.61]*
end	teacher	+52.10*	+23.95*	+18.80*	+13.92*	+11.44*	+0.75	-2.44	-7.76 [-9.07, -6.45]*
	loss term	-5.21	+15.84*	+6.33	-0.65	-6.00	-1.20	+1.55	-0.93 [-2.27, +0.41]
	total	+46.89*	+39.79*	+25.13*	+13.27*	+5.44	-0.45	-0.89	-8.70 [-10.07, -7.33]*
	abrupt — none	-14.82*	-21.92*	-36.58*	-48.44*	-56.27*	-62.16*	-62.60*	-8.70 [-10.07, -7.33]*

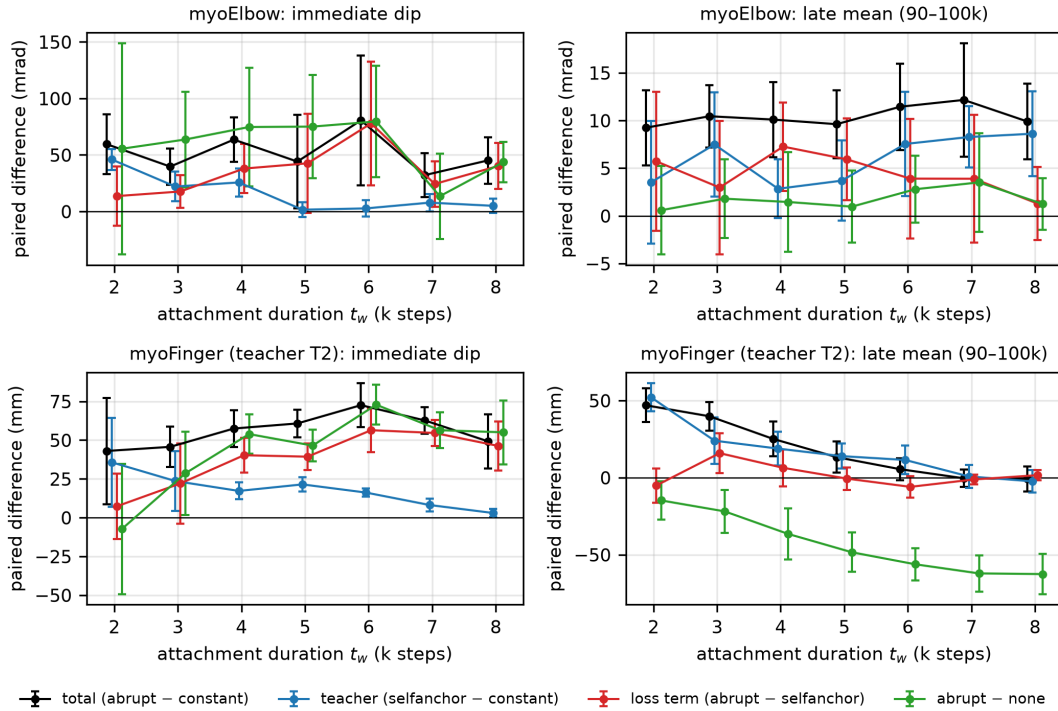


Fig. 4. What the drop at withdrawal is made of, against attachment duration. At the immediate dip (left) the teacher component (blue) shrinks the longer the teacher was attached, while the loss-term component (red) does not. At the late mean (right) the withdrawn learner is level with the never-guided one on the elbow (green, top) and far ahead of it on the finger (green, bottom). Paired means with 95% intervals, $n = 12$; elbow in mrad, finger in mm (teacher T2).

is kept on one of three bodies. On the elbow neither kind of help is worth much and neither is kept, but neither leaves the learner worse off than having had no help at all.

Dependence. No withdrawn learner, on either MyoSuite body and in either regime, ends detectably behind one that was never guided, and where the teacher is worth enough to bound the difference, the bound lies well below that worth; on the elbow, training to 200k narrows it to a quarter of the teacher's worth. The setting differs from the studies that motivate the concern: attachment duration rather than feedback frequency,

and assessment after long continued practice rather than a no-practice retention test [14]. Human studies do find withdrawal costs at retention after fixed practice [13], and a field study finds students whose tutor was withdrawn below students who never had one [69]. After long practice a weaned learner can end above a from-scratch one [6], and a regulariser removed after the early phase costs little [70], though early experience can also leave lasting marks on a deep reinforcement learner [71]. Here early guidance leaves a head start rather than a handicap.

TABLE VI

TAKING THE BODY MODEL AWAY AT 8K, LATE MEAN. WORTH: KEEP (THE MODEL NEVER WITHDRAWN) AGAINST BLANK. EACH WITHDRAWN CONDITION IS COMPARED WITH THE NEVER-HAD CONDITION THAT TRAINS AS IT DOES AFTER WITHDRAWAL. SHORTFALL: AGAINST KEEPING THE MODEL. PART OF THE BENEFIT SURVIVES ON THE FINGER ONLY. PAIRED MEANS; * MARKS AN INTERVAL EXCLUDING ZERO.

	worth	withdrawn — never had		shortfall
	keep—blank	purge—blank	matched—blank64	matched—keep
myoElbow (mrad), $n=12$	−3.90*	+0.35	−0.69	+7.13*
myoFinger (mm), $n=12$	−46.19*	−26.05*	−11.44*	+14.46*
4-joint arm (mm), $n=24$	−94.99*	+5.84	+10.71	+45.99*

Practical consequences. The drop that follows withdrawal is the objective changing under the learner, and a term toward the learner’s own snapshot removes it. Pipelines that wean a student off distillation [6], [7] can therefore taper or re-anchor the term rather than cut it, without giving up what the teacher taught. More generally, a withdrawal that changes two things at once cannot be read as the cost of the thing withdrawn: the teacher needs a self-anchor to separate its knowledge from its loss term, an imitation-only learner a gateon condition to separate the teacher from the first use of its own gradient, and the body model a never-had reference with the same real data per update, without which a smaller batch reads as a retained benefit.

VI. LIMITATIONS AND FUTURE WORK

The study is single-task reach in simulation, on deterministic, fully observed bodies, which favour a frozen forward model; the one-joint bodies’ held-out targets interpolate between the training targets, and the imagined reward uses exact forward kinematics. Partially observed, stochastic bodies and a learned reward are the natural next test of the body model. The finger’s competent teacher is the best of twelve candidate runs; T1 shows that a teacher that barely moves gives the opposite result, and a graded series of teachers would map worth against competence. The self-anchor holds what the student has absorbed by withdrawal, so the fall of the teacher component with attachment measures absorption by construction; an anchor taken at a fixed earlier time would test it independently. Withdrawal is a single step, attachment varies from 2k to 8k steps, contrasts were not pre-registered, the endpoint was changed after an internal review, and several conditions are still moving at 100k, among them the four-joint arm’s withdrawn learner; only the elbow is trained to 200k. Gradual fading, longer training on the harder bodies and a physical robot are left to future work.

VII. CONCLUSION

A body model and a teacher are given to a muscle-driven learner, controlled with a random counterpart, and taken away. A frozen babble-trained body model helps on all six bodies, more the more joints the body has, and its benefit lasts mainly while its imagination runs. A teacher is worth its competence, and once withdrawn it leaves no learner detectably worse than one never taught; on the finger most of what it taught stays,

and the drop at withdrawal is the change of objective, not forgetting.

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